TREE OF THOUGHTS

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Deliberate Problem Solving with Large Language Models | NeurIPS 2023

Turn language-model inference into deliberate search over coherent intermediate thoughts: generate alternatives, evaluate partial solutions, keep promising branches and backtrack when needed.

deliberate search · planning · self-evaluation

1 Problem

- IO and CoT decoding commit token by token; an early bad choice can trap the remainder of the solution.
- Self-consistency explores complete chains but cannot branch, compare or repair locally within a chain.
- Planning tasks need lookahead, strategic exploration and backtracking—an LM analogue of deliberate System 2 reasoning.

2 State space

THOUGHTS AS SEARCH NODES

- A state contains the input plus thoughts chosen so far; a thought is a coherent unit sized for the problem.
- Thoughts can be equations in Game of 24, paragraph plans in writing or candidate words in crosswords.
- Good decomposition must be small enough for diverse generation yet large enough for meaningful evaluation.

3 Generate & evaluate

- Generate candidates independently for open spaces or sequentially in one propose prompt for constrained spaces.
- Evaluate each state as a value (sure/maybe/impossible or a score) or compare states through repeated voting.
- The same frozen LM supplies both ideas and heuristics; no reward model or additional training is required.

4 Search

- Breadth-first search keeps the top b states at each shallow step; used for Game of 24 and creative writing.
- Depth-first search expands the most promising state, prunes impossible subtrees and backtracks; used for mini crosswords.
- Components are modular: thought granularity, generator, evaluator, LM and search algorithm can change independently.

5 Evidence

- Game of 24: GPT-4 IO 7.3%, CoT 4%, CoT-SC 9%, versus ToT 45% at b=1 and 74% at b=5.
- Creative-writing coherence rises from 6.19 IO / 6.93 CoT to 7.56 ToT; humans prefer ToT over CoT 41 versus 21 times.
- Mini crosswords: word success rises below 16% to 60%, and ToT solves 4/20 games; removing backtracking drops word success to 20%.

6 Limits & meaning

GENERAL, BUT EXPENSIVE

- ToT is unnecessary when direct prompting already works and was tested on only three deliberately hard tasks.
- Multiple generation and evaluation calls cost far more than CoT; Game of 24 uses about 5.5k completion tokens per case.
- Imperfect self-evaluation can prune correct paths, but the framework connects LMs with classical BFS, DFS, A* and MCTS ideas.